BASE CHANGE FOR THE DERIVED PUSHFORWARD UNDER TOR INDEPENDENCE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- For any object of the derived category on X there is a canonical base change map Lg^*Rf_*E to Rf'_*L(g')^*E attached to the cartesian diagram. used at proof lines 5379-5384. Supplies the arrow whose invertibility the lemma asserts.
- For an affine morphism the pushforward of a derived tensor with the pullback of the structure sheaf is the derived tensor of the pushforward, and for an affine base change the pushforward of the structure sheaf of the fibre product is the pullback of the pushforward of the structure sheaf. invoked at proof lines 5398-5407. Rewrites the pushed-forward comparison map into a map of derived tensors on S, reducing the assertion to Lf^*g_*O_{S'} = f^*g_*O_{S'}.
- Tor independence of two rings over a base means all higher Tor vanish, equivalently the derived tensor product agrees with the ordinary one. invoked at proof lines 5410-5417. Converts the Tor independence hypothesis into the equality Lf^*g_*O_{S'} = f^*g_*O_{S'} that finishes the proof.

Source reference. Original remark 08HY: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 5380-5384 for the canonical map; checking its invertibility is inside the counted span.

Source reference. Original lemma 08I8: sources/stacks-perfect/pinned-original-source.tex, source lines 867–867. Cited at proof lines 5394-5396; the rewriting of the pushed-forward comparison map is inside the counted span.

Source reference. Original lemma 08DW: sources/stacks-perfect/pinned-original-source.tex, source lines 581–581. Cited at proof lines 5396-5397 and again at line 5417; both uses (quasi-coherent cohomology sheaves and Lf^* = f^* on g_*O_{S'}) are inside the counted span.

Source reference. Original lemma 08DX: sources/stacks-perfect/pinned-original-source.tex, source lines 620–620. Cited at proof lines 5396-5397 to see Lg^*Rf_*E

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has quasi-coherent cohomology sheaves; the check enabling Lemma affine-morphism is inside the counted span.

Source reference. Original lemma 08D5: sources/stacks-perfect/pinned-original-source.tex, source lines 661–661. Cited at proof lines 5396-5397 to see Rf'_*L(g')^*E has quasi-coherent cohomology sheaves; the check enabling Lemma affine-morphism is inside the counted span.

Source reference. Original lemma 01SD: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 5397-5398 to note g' is affine; the use of the affine pull-push formula for g' is inside the counted span.

Source reference. Original lemma 08I9: sources/stacks-perfect/pinned-original-source.tex, source lines 888–888. Cited at proof lines 5398-5402 to rewrite the map as a map of derived tensors on S; the resulting reduction is inside the counted span.

Source reference. Original lemma 02KG: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 5403-5406 for g'_*O_{X'} = f^*g_*O_{S'}; the identification and its use are inside the counted span.

Source reference. Original lemma 08EU: sources/stacks-perfect/pinned-original-source.tex, source lines 5269–5269. Cited at proof lines 5406-5408 to reduce to Lf^*g_*O_{S'} = f^*g_*O_{S'}; the verification of that equality is inside the counted span.

Source reference. Original lemma 08HX: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 5410-5414 to conclude Tor_i^R(A, R') = 0 for i > 0; the derived-tensor identification finishing the proof is inside the counted span.

Lemma 0.2. *Let $g : S' \to S$ be a morphism of schemes. Let $f : X \to S$ be quasi-compact and quasi-separated. Consider the base change diagram*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array}$$

If X and S' are Tor independent over S, then for all $E \in D_{QCoh}(\mathcal{O}_X)$ the canonical arrow $Lg^*Rf_*E \to Rf'_*L(g')^*E$ is an isomorphism.

Proof. For any object E of $D(\mathcal{O}_X)$ we can use Cohomology, Remark 08HY to get a canonical base change map $Lg^*Rf_*E \to Rf'_*L(g')^*E$. To check this is an isomorphism we may work locally on S'. Hence we may assume $g : S' \to S$ is a morphism of affine schemes. In particular, g is affine and it suffices to show that

$$Rg_*Lg^*Rf_*E \to Rg_*Rf'_*L(g')^*E = Rf_*(Rg'_*L(g')^*E)$$

is an isomorphism, see Lemma 08I8 (and use Lemmas 08DW, 08DX, and 08D5 to see that the objects $Rf'_*L(g')^*E$ and Lg^*Rf_*E have quasi-coherent cohomology sheaves). Note that g' is affine as well (Morphisms, Lemma 01SD). By Lemma 08I9 the map becomes a map

$$Rf_*E \otimes^{\mathbf{L}}_{\mathcal{O}_S} g_*\mathcal{O}_{S'} \longrightarrow Rf_*(E \otimes^{\mathbf{L}}_{\mathcal{O}_X} g'_*\mathcal{O}_{X'})$$

Observe that $g'_*\mathcal{O}_{X'} = f^*g_*\mathcal{O}_{S'}$ (by affine base change, see Cohomology of Schemes, Lemma 02KG). Thus by Lemma 08EU it suffices to prove that $Lf^*g_*\mathcal{O}_{S'} = f^*g_*\mathcal{O}_{S'}$.

This follows from our assumption that X and S' are Tor independent over S. Namely, to check it we may work locally on X, hence we may also assume X is affine. Say $X = \mathrm{Spec}(A)$, $S = \mathrm{Spec}(R)$ and $S' = \mathrm{Spec}(R')$. Our assumption implies that A and R' are Tor independent over R (More on Algebra, Lemma 08HX), i.e., $\mathrm{Tor}_i^R(A, R') = 0$ for $i > 0$. In other words $A \otimes^{\mathbf{L}}_R R' = A \otimes_R R'$ which exactly means that $Lf^* g_* \mathcal{O}_{S'} = f^* g_* \mathcal{O}_{S'}$ (use Lemma 08DW). $\square$